

Assignment: Counting Chocolates with Computer Vision

Background

You are given a photograph of a box of chocolates. Your task is to write a Python program using OpenCV that **automatically counts how many chocolates are in the image**.

This is a real problem in industrial settings, automated counting of objects on a production line, quality control, inventory management. Your solution does not need to be perfect, but it needs to be justified and reasoned.

The Image

Download `img_3.jpg` from Canvas before getting started.



`img_3.jpg` — download this file from Canvas

Load the image `img_3.jpg`. Take a close look at it:

- What do the chocolates look like compared to the background?
- Are they darker or lighter than their surroundings?
- Are they roughly the same size?
- Are they uniform in appearance?

These observations will drive your strategy.

Your Task

Write a Python script that:

1. Loads the image and converts it to grayscale
 2. Uses a strategy to isolate the chocolates from the background
 3. Counts the number of chocolates detected
 4. Draws a marker on each detected chocolate and labels it with its number
 5. Prints the final count to the terminal
-

Questions to Guide Your Strategy

Before writing any code, answer the following questions:

1. How do the chocolates differ from the background in terms of pixel intensity?
 2. If you apply a threshold to the grayscale image, what threshold value separates the chocolates from the background? How did you determine it?
 3. How can you use the expected size of a chocolate to your advantage?
 4. What happens if two chocolates are close together — how might that affect your count?
 5. What does it mean for the pixels inside a candidate region to be "uniform"? Why is uniformity a useful property to check?
-

Getting Started

Python — getting_started.py

```
import cv2
import numpy as np

img = cv2.imread("img_3.jpg")
cv2.imshow("Original", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Start by just displaying the image. Study it carefully before writing any detection logic.